

Material Test Report / Certificate EC-HNBR-B25-0339

ACT Elastomer Lab (compound by Permian Polymer Compounding, Odessa TX) | ACT Burleson - Lab Bldg 3

Certificate No. / Date	EC-HNBR-B25-0339 / 2025-05-21
Heat / Batch No.	B25-0339
Customer	Internal - production release
Purchase Order	PO-88490 (compound purchase)
Product	HNBR compound ACT-H90, 90 Shore A, molded packer/plug elements & o-rings
Specification(s)	ACT-ES-05 Rev D (element compounds); batch release per ASTM D2000 line callout
Quantity	Batch B25-0339 / 2,200 lb compound

Chemical Composition (formulation)

Element	Value	Element	Value
Polymer	HNBR, 96% saturation	Carbon black	N550, 65 phr
ACN content	36 %	Plasticizer	TOTM, 5 phr
Cure system	Peroxide	Antioxidant	ZMTI/CDPA, 2 phr

Mechanical / Physical Properties

Property	Result	Requirement
Hardness	91 Shore A	90 +/- 3
Tensile strength	3,420 psi	>= 2,900 psi
Elongation at break	185 %	>= 150 %
100% Modulus	1,850 psi	>= 1,500 psi
Compression set, 70 hr @ 302 degF	18 %	<= 25 %
Air aged 70 hr @ 302 degF, hardness change	+4 pts	+/- 8 pts max
RGD resistance (NORSOK M-710 Annex B)	Rating 2222	<= 3333
Fluid immersion, IRM 903, 70 hr @ 302 degF, volume swell	+7.2 %	<= 12 %

Thermal Treatment

Post-cure 4 hr @ 320 degF (molded articles)

NDE / Additional Testing

Visual per RMA Class 2; durometer mapped 5 points per article

Sour Service (NACE MR0175 / ISO 15156) Statement

Suitable for sour service exposure per ACT-ES-05; H2S resistance verified by immersion aging.

Remarks

Release covers molded elements ACT-PKR-4500-03B (center element), ACT-FP-5500-02 frac plug elements, and ACT-ORG series HNBR o-rings produced from batch B25-0339. Cure date code 2521. Shelf life 5 years from cure per ARP5316.

Certification

We hereby certify that the material described above was manufactured, sampled, tested and inspected in accordance with the referenced specifications and was found to meet those requirements. This certificate is issued in accordance with EN 10204 Type 3.1.

Role	Name	Signature	Date
Authorized Signatory	K. Patel, Metallurgist / Materials Lab Supervisor, ACT	/signed electronically/	2025-05-21
ACT Receiving QA Review	S. Nguyen	/signed electronically/	2025-05-21